

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Leak Test Kit, Part Number 2892320
- Fluorescent dye additive for fuel and lubricating oil, Part Number 3376891
- Fluorescent dye additive for coolant, Part Number 3377438

Additional Service Items

- No additional service items required.

General Information

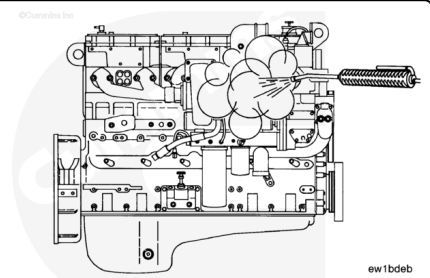
The Fluorescent Tracer Dye Test is used to aid in locating fluid leaks in the lubricating oil system, cooling system, and fuel system. The instructions and illustrations in this procedure are generic and do **not** cover component specific instructions. For component specific testing instructions, see the appropriate procedure in the corresponding Cummins® Service Manual or the equipment manufacturer service information.

Preparatory Steps

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠



When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

⚠ CAUTION ⚠

Protect all electrical components, openings, and wiring from the full force of the cleaner spray nozzle.

Some rust prevention oils applied at the factory have fluorescent dyes, which can provide false test results. Check suspect areas with a black light to make sure there is not already fluorescent dye present before adding fluorescent dyes to the fluid under test.

Steam clean the affected areas on the engine.

If steam is not available, use an appropriate solvent to clean the engine.

Setup

Initial Setup

See the corresponding Cummins® Service Manual or the equipment manufacturer service information to determine the total volume of the fluid system being tested. Select the appropriate fluorescent dye additive for the fluid system being tested.

- For lubricating oil or fuel use fluorescent dye additive, Part Number 3376891.
- For coolant use fluorescent dye additive, Part Number 3377438.

Determine the amount of fluorescent dye additive required for the fluid system being tested. Use the instructions provided with the fluorescent dye additive being used.

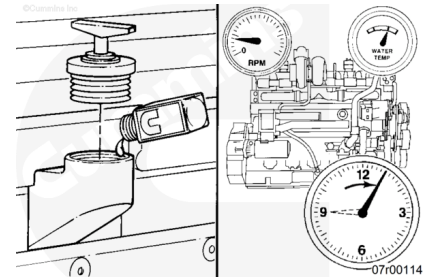
Lubricating Oil

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.



Remove the lubricating oil fill cap and add the appropriate amount of fluorescent dye additive to the lubricating oil system. Install the lubricating oil fill cap.

Note : Lubricating oil with high levels of soot may need to be changed in order to make the fluorescent dye visible. Up to three times the standard amount of fluorescent dye additive may be used if necessary. The use of fluorescent dye additive does not interfere with engine performance and can be left in the lubricating oil after use.

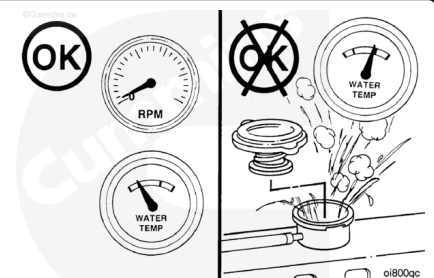
Operate the engine in a manner to distribute the fluorescent dye additive throughout the lubricating oil system evenly. This can typically be achieved by operating the engine at high idle for approximately five minutes.

For engine or equipment specific test procedures, see the corresponding Cummins® Service Manual or the equipment manufacturer's service information.

Coolant

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

The coolant line(s) can be hot and under pressure. Do not disconnect the coolant lines while the engine is running or before the system has cooled down after engine shutdown. Heated coolant spray can cause personal injury. To reduce the possibility of personal injury, use eye and face protection, gloves and appropriate personal protective equipment.

Verify the coolant temperature is below 50°C [120°F].

Remove the radiator cap.

Add the appropriate amount of fluorescent dye additive to the cooling system.

Install the radiator cap.

Operate the engine in a manner to evenly distribute the fluorescent dye additive throughout the cooling system. Raise coolant temperature until coolant thermostats are open. This can typically be achieved by operating the engine at high idle while monitoring engine coolant temperature. Operate engine for an additional ten minutes after coolant thermostats have opened.

For coolant thermostat opening temperatures, see corresponding Cummins® Service Manual or equipment manufacturer service information.

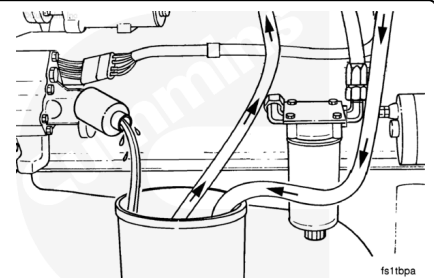
Fuel

⚠ WARNING ⚠

Fuel can be returned at highly elevated temperatures. Wear safety glasses and protective gloves and clothing when performing this test. Avoid any contact with returned fuel.

⚠ WARNING ⚠

Fuel is flammable. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas sharing ventilation to reduce the possibility of severe personal injury or death when working on the fuel system.



Note : This section only applies to engines operating on diesel fuel.

Test method when operating engine from equipment fuel reservoir:

Remove fuel fill cap and add appropriate amount of fluorescent dye additive to fuel tank. Install fuel fill cap.

Operate equipment to evenly distribute fluorescent dye additive in fuel system.

Test method when operating engine from a temporary fuel reservoir:

Fill a container suitable for fuel with clean diesel fuel. Add appropriate amount of fluorescent dye additive to container. Mix fuel and fluorescent dye additive for even distribution.

Disconnect fuel supply and return lines from engine. See appropriate procedures in corresponding Cummins® Service Manual. Fabricate temporary fuel lines to connect from temporary fuel reservoir to engine fuel inlet and return ports.

Operate equipment to evenly distribute fluorescent dye additive in fuel system.

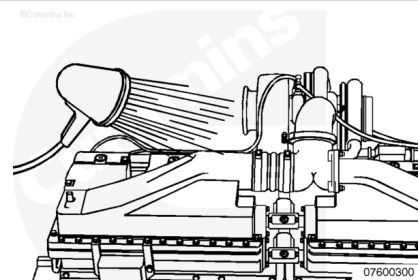
Test

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

To reduce the possibility of personal injury, keep hands, long hair, jewelry, and loose-fitting or torn clothing away from fans and other moving parts.



Inspect area of suspected leak for presence of fluorescent dye. Use the black light from leak test kit, Part Number 2892320.

Operation of the engine may be necessary during the inspection process to determine the origin of the leak.

Always start at the bottom of the engine or component being inspected and work toward the top. Liquids will flow to the lowest point before forming droplets. Follow the fluorescent dye trail upward to determine the origin of the leak. If a leak is suspected in a given area, verify the fluorescent dye trail does **not** originate from a location above.

Components like the cooling fan may create air flow that will distribute the fluorescent dye in a horizontal direction. In these cases, follow the fluorescent dye trail in the opposite direction of flow to find the origin of the leak.

Rotating components like the flywheel or vibration damper, will distribute fluorescent dye in a circular pattern. For this situation, determine if the fluorescent dye is being dripped on the component from above and being redistributed, or if the leak originates at the seal near the center of the component.

Repair or replace the component or components that the fluorescent dye originates from.

If the origin of the fluorescent dye can **not** be found, consider cleaning the engine or component and test again.

If fluorescent dye is **not** found during the inspection, no leak is present.

Finishing Steps

Install any components that were removed for access during the test.

To verify repair:

- Operate equipment in same manner used to generate leak(s).
- Check for leaks.